

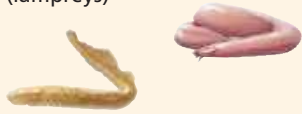
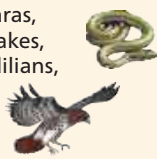
34 Chapter Review



➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkz9t.

Key Concept	Clade	Description
CONCEPT 34.1 Chordates have a notochord and a dorsal, hollow nerve cord (pp. 719–722) ? Describe likely features of the common ancestor of chordates. Explain your answer.	Cephalochordata (lancelets)  Urochordata (tunicates) 	Basal chordates; marine suspension feeders that exhibit four key derived characters of chordates Marine suspension feeders; larvae display the derived traits of chordates
CONCEPT 34.2 Vertebrates are chordates that have a backbone (pp. 722–725) ? Identify the shared features of early fossil vertebrates.	Myxini (hagfishes) and Petromyzontida (lampreys) 	Jawless aquatic vertebrates with reduced vertebrae; hagfishes have a head with a skull and brain, eyes, and other sensory organs; some lampreys feed by attaching to a live fish and ingesting its blood
CONCEPT 34.3 Gnathostomes are vertebrates that have jaws (pp. 725–730) ? How would the appearance of organisms with jaws have altered ecological interactions? Provide supporting evidence.	Chondrichthyes (sharks, rays, skates, ratfishes)  Actinopterygii (ray-finned fishes)  Actinistia (coelacanths)  Dipnoi (lungfishes) 	Aquatic gnathostomes; have a cartilaginous skeleton, a derived trait formed by the reduction of an ancestral mineralized skeleton Aquatic gnathostomes; have a bony skeleton and maneuverable fins supported by rays Ancient lineage of aquatic lobe-fins still surviving in Indian Ocean Freshwater lobe-fins with both lungs and gills; sister group of tetrapods
CONCEPT 34.4 Tetrapods are gnathostomes that have limbs (pp. 730–734) ? Which features of amphibians restrict most species to living in aquatic or moist terrestrial habitats?	Amphibia (salamanders, frogs, caecilians) 	Have four limbs descended from modified fins; most have moist skin that functions in gas exchange; many live both in water (as larvae) and on land (as adults)
CONCEPT 34.5 Amniotes are tetrapods that have a terrestrially adapted egg (pp. 734–741) ? Explain why birds are considered reptiles.	Reptilia (tuataras, lizards and snakes, turtles, crocodilians, birds) 	One of two groups of living amniotes; have amniotic eggs and rib cage ventilation, key adaptations for life on land
CONCEPT 34.6 Mammals are amniotes that have hair and produce milk (pp. 741–747) ? Describe the origin and early evolution of mammals.	Mammalia (monotremes, marsupials, eutherians) 	Evolved from synapsid ancestors; include egg-laying monotremes (echidnas, platypus); pouched marsupials (such as kangaroos, opossums); and eutherians (placental mammals, such as rodents, primates)